Randomization

28/02/2024

Description

This file enables concatenation and randomization of sign strings which occur on different parts of an object (indicated by underscore '_'). We do not know the order of these strings (if any), hence, if we want to combine them, they need to be randomized.

Session Info

Give the session info (reduced).

```
## [1] "R version 4.3.2 (2023-10-31)"
```

```
## [1] "x86_64-pc-linux-gnu"
```

Load libraries

If the libraries are not installed yet, you need to install them using, for example, the command: install.packages("ggplot2").

```
#library(stringr)
```

Load Data

```
sign.data <- read.csv("~/Github/UpperPalCombinatorics/data/signBase/signBase_clean.csv")
head(sign.data)
```

```
##   object_id ensemble_id archaeological_unit                site_name location
## 1   sgr0074  sgr_ens_01  Middle Magdalenian Saint-Germain-la-Rivière       NA
## 2   sgr0073  sgr_ens_01  Middle Magdalenian Saint-Germain-la-Rivière       NA
## 3   sgr0072  sgr_ens_01  Middle Magdalenian Saint-Germain-la-Rivière       NA
## 4   sgr0070  sgr_ens_01  Middle Magdalenian Saint-Germain-la-Rivière       NA
## 5   sgr0067  sgr_ens_01  Middle Magdalenian Saint-Germain-la-Rivière       NA
## 6   sgr0062  sgr_ens_01  Middle Magdalenian Saint-Germain-la-Rivière       NA
##    layer         dating_method  date_bp_min min_lab_number date_bp_max
## 1 burial AMS, 858 (GifA 95456) 15,780 ± 200             NA          NA
## 2 burial AMS, 858 (GifA 95456) 15,780 ± 200             NA          NA
## 3 burial AMS, 858 (GifA 95456) 15,780 ± 200             NA          NA
## 4 burial AMS, 858 (GifA 95456) 15,780 ± 200             NA          NA
## 5 burial AMS, 858 (GifA 95456) 15,780 ± 200             NA          NA
## 6 burial AMS, 858 (GifA 95456) 15,780 ± 200             NA          NA
```

```
##   max_lab_number excavation_year              material
## 1             NA            1934 animal tooth, red deer
## 2             NA            1934 animal tooth, red deer
## 3             NA            1934 animal tooth, red deer
## 4             NA            1934 animal tooth, red deer
## 5             NA            1934 animal tooth, red deer
## 6             NA            1934 animal tooth, red deer
##                 object_type length_mm width_mm depth_mm preservation
## 1 personal ornament, pendant        NA       NA       NA     complete
## 2 personal ornament, pendant        NA       NA       NA     complete
## 3 personal ornament, pendant        NA       NA       NA     complete
## 4 personal ornament, pendant        NA       NA       NA  fragmentary
## 5 personal ornament, pendant        NA       NA       NA  fragmentary
## 6 personal ornament, pendant        NA       NA       NA     complete
##   sign_coding coding_certainty coding_clean
## 1        |||X          Certain         |||X
## 2          ||          Certain           ||
## 3          XX   Very uncertain           XX
## 4       |||||          Certain        |||||
## 5          ||   Very uncertain           ||
## 6        ||1|          Certain         ||1|
```

Randomize

Randomize sign sequences per object.

```r
# set seed to get reproducible result
set.seed(09012020)
# initialize empty vector
coding.random <- c()
# run loop to randomize, and add new column to data frame
for (i in 1:nrow(sign.data)){
  # use single coding per row
  coding.local <- sign.data[i, ]$coding_clean
  # split on underscores surrounded by white spaces
  coding.split <- unlist(strsplit(as.character(coding.local), split = " _ "))
  # randomize order
  coding.random.local <- sample(coding.split)
  # concatenate with underscore and white space again
  coding.random.local <- paste(coding.random.local, collapse = " _ ")
  # add to vector
  coding.random <- c(coding.random, coding.random.local)
}

# add column with randomized order to data frame
sign.data$coding_random <- coding.random

# safe to file
write.csv(sign.data, "~/Github/UpperPalCombinatorics/data/signBase/signBase_randomized.csv", row.names =
```